

From Accuracy to Explainability: Understanding CNN Errors with xAI

xAI-Proj-B Course Project – CNN-based Image Classification

Course Context & Motivation

Context

- Explainable AI course
- All teams: first CNN + xAI project

Motivation

- CNN accuracy can look convincing
- But:
 - Are errors random?
 - Can we trust confident predictions?

CNN Basics (Learner Perspective)

CNN Intuition

- CNNs learn hierarchical visual features
- Early layers: edges, textures
- Deeper layers: shapes, object parts

Why CNNs

- Designed for image data
- Widely used in object classification

Model Choice & Baselines

Our Setup

- Backbone: ResNet18
- 10-class image classification

Baseline Design

- Different pretraining sources
- Different training strategies

Research Question & Red Thread

Research Question

- What kinds of errors does a CNN make?
- How confident is it when it is wrong?
- Can xAI help us understand these errors?

Red Thread

- Build → Observe → Question → Explain

Initial Results: Accuracy Is Not Enough

Observation

- Models reach reasonable accuracy
- Model B performs best overall

But

- Accuracy alone does not show:
 - Error structure
 - Confidence behavior

Why Sample-Level Analysis?

Limitation of Aggregate Metrics

- Accuracy averages over samples
- Risky errors are hidden

Our Idea

- Look at individual predictions
- Combine:
 - correctness
 - confidence
 - loss

Difficulty Table & Error Subsets

Per-sample Metrics

- Prediction & true label
- Confidence
- Loss

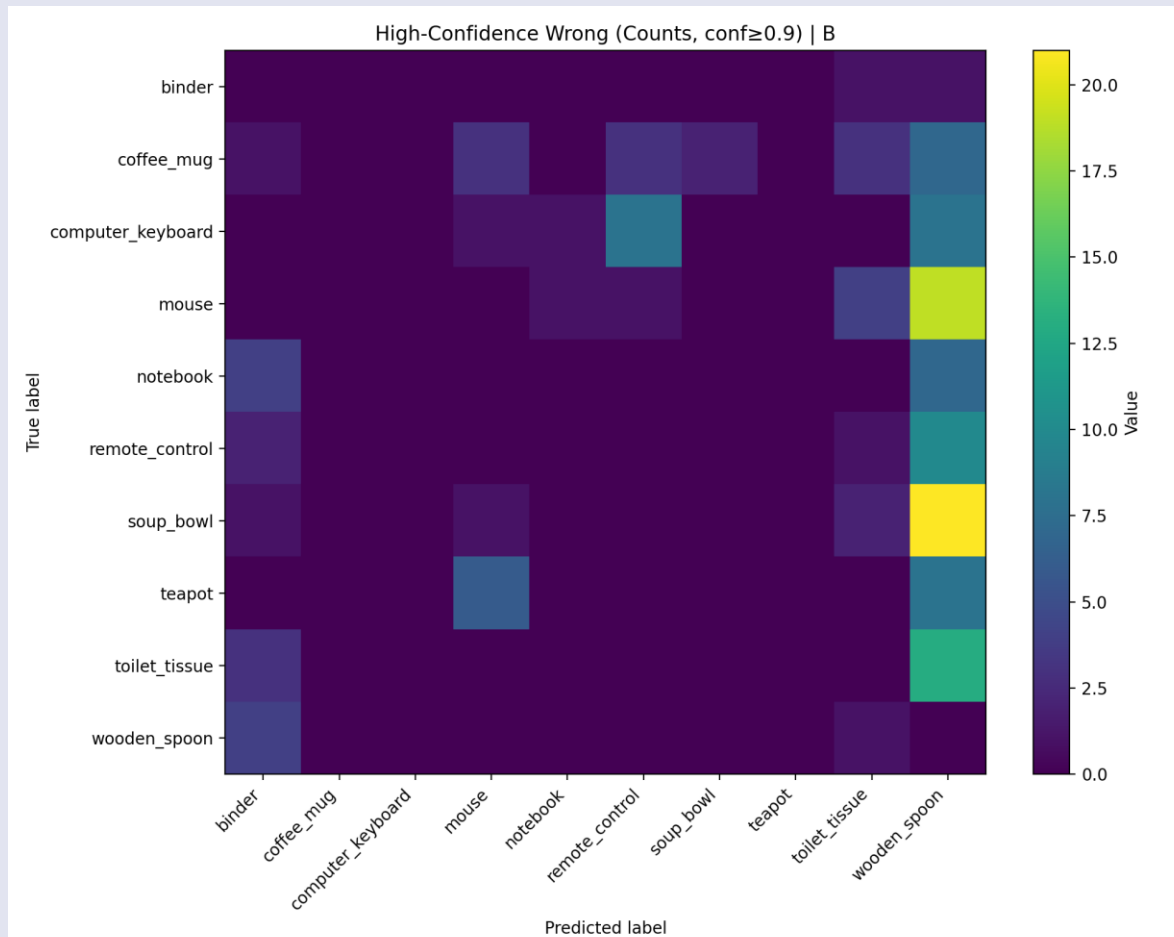
Defined Subsets

- Easiest correct
- Hardest correct
- Low-confidence correct
- **High-confidence wrong**

Top conf wrong (first 5):

	true_label	pred_label	confidence	loss	top2_margin
2985	soup_bowl	wooden_spoon	0.998722	8.548472	0.998160
2571	remote_control	wooden_spoon	0.994697	6.331002	0.992916
527	coffee_mug	remote_control	0.992541	7.117701	0.990629
2829	soup_bowl	wooden_spoon	0.992233	6.365205	0.988245
3906	toilet_tissue	wooden_spoon	0.991626	6.058759	0.988923

Focused confusion on high-confidence wrong



Interim Summary: What We Learned So Far

So Far

- Errors are not random
- Confidence does not guarantee correctness
- Some classes dominate errors

Why Explainable AI?

What is Grad-CAM?

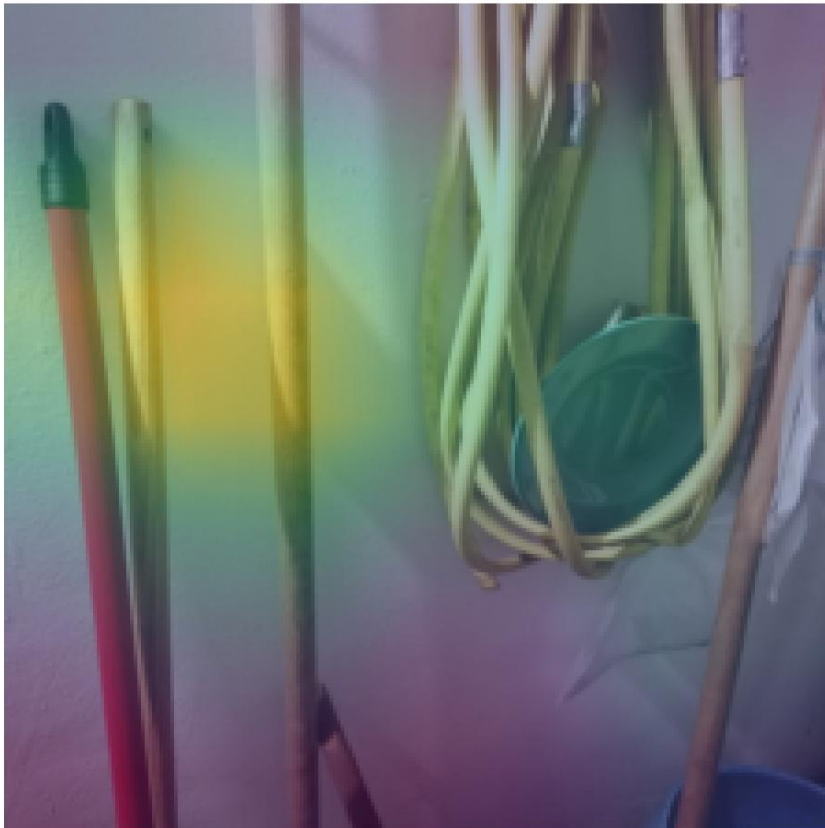
- A visual explanation method for CNNs
- Highlights image regions that contribute most to a specific prediction
- Produces a heatmap over the input image

Method (Our Use Case)

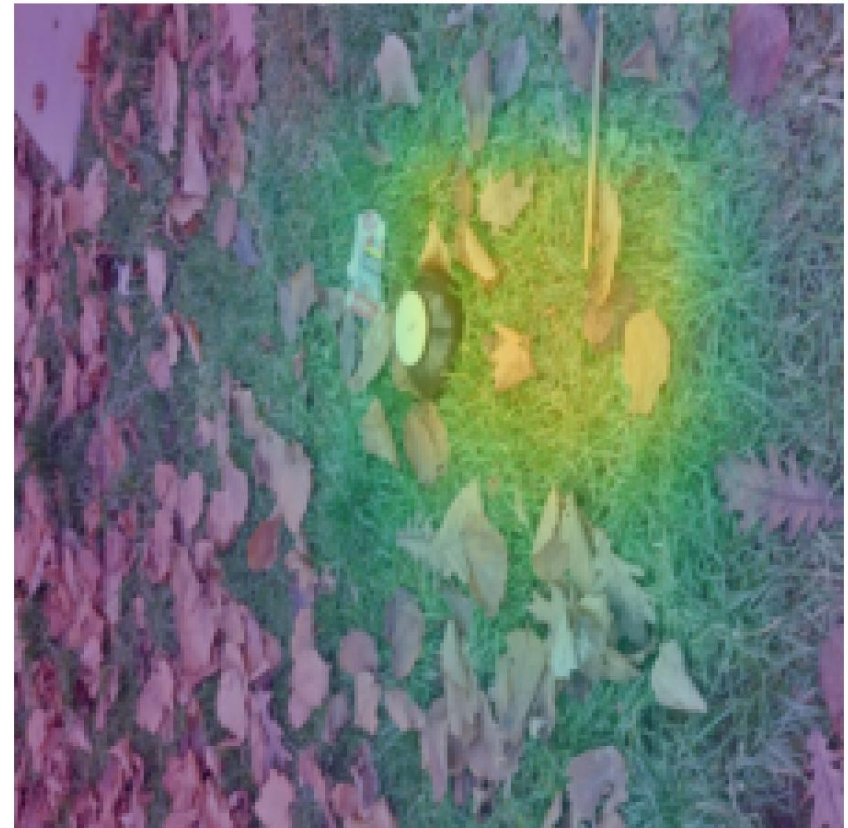
- Apply Grad-CAM to **high-confidence wrong samples**
- Analyze whether the model focuses on object regions or spurious cues

Grad-CAM Examples (1): soup_bowl → wooden_spoon

true=soup_bowl | pred=wooden_spoon | conf=0.999

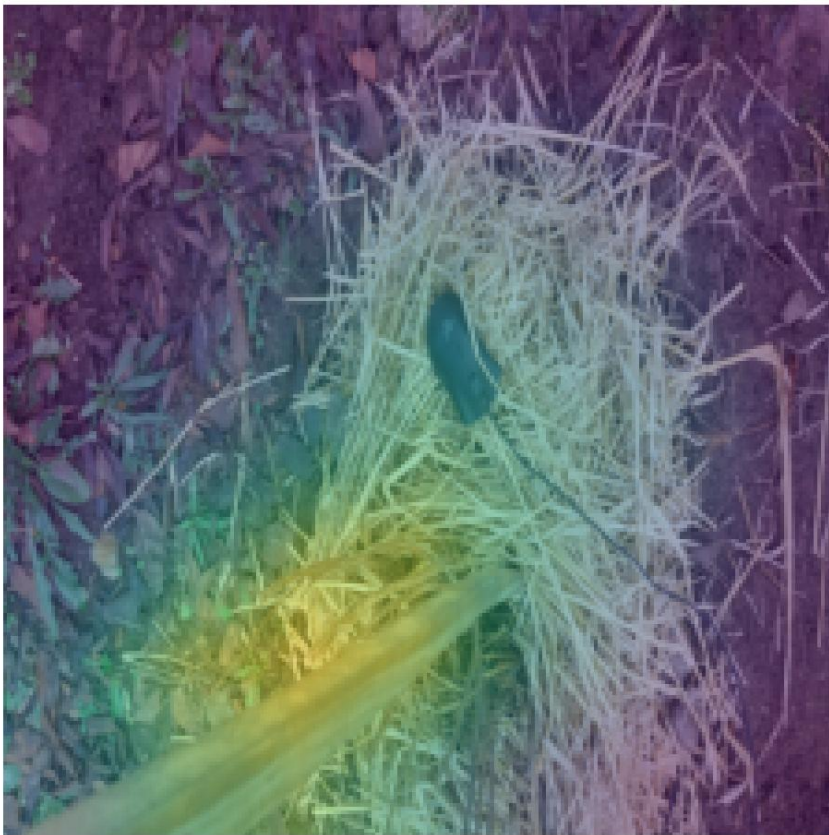


true=soup_bowl | pred=wooden_spoon | conf=0.992

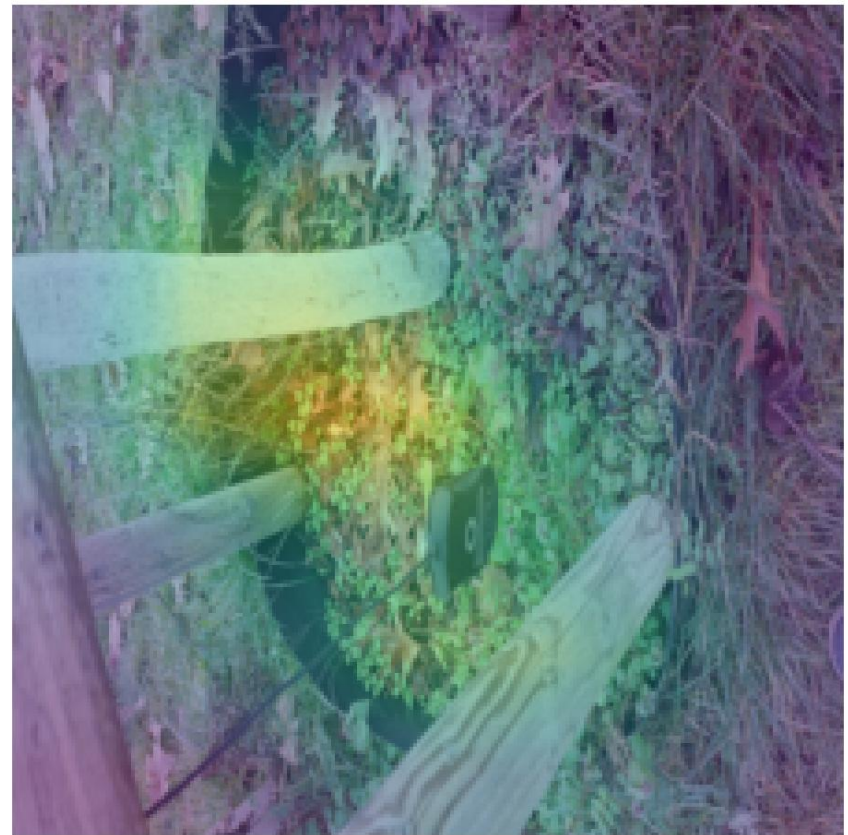


Grad-CAM Examples (2): mouse → wooden_spoon

true=mouse | pred=wooden_spoon | conf=0.989



true=mouse | pred=wooden_spoon | conf=0.986

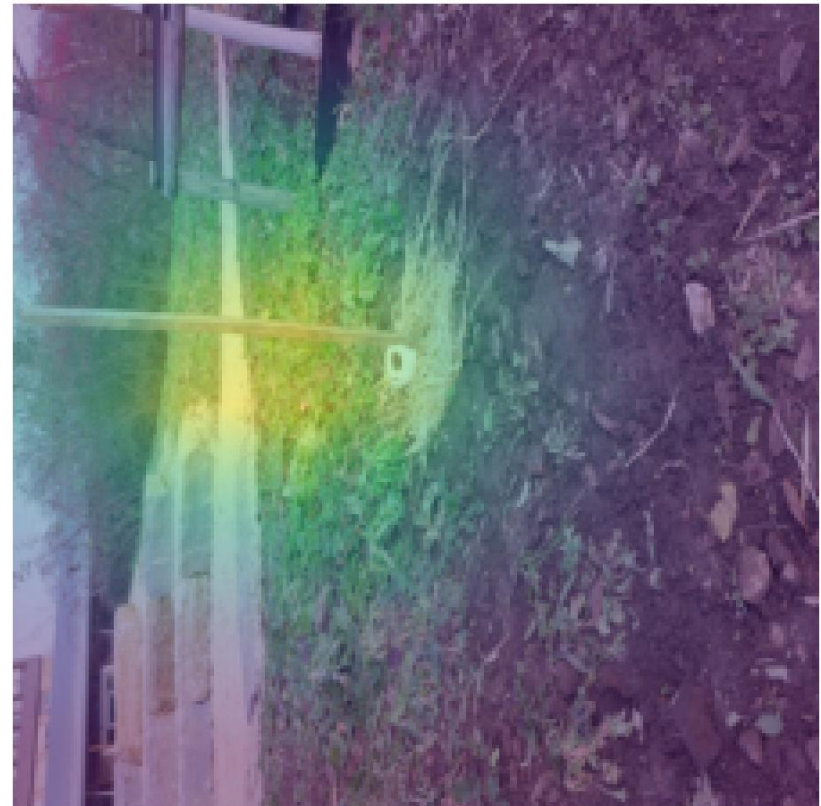


Grad-CAM Examples (3): toilet-tissue → wooden_spoon

true=toilet_tissue | pred=wooden_spoon | conf=0.992



true=toilet_tissue | pred=wooden_spoon | conf=0.983



Grad-CAM Insights

Observation

- Attention focuses on:
 - wooden textures
 - elongated shapes
 - background regions

Interpretation

- Shortcut / texture bias
- Overgeneralization of wooden_spoon features

Reflection, Trust & Next Steps

Reflection

- Accuracy $\neq$ reliability
- Explainability changes how we trust models

Next Steps

- Cross-model comparison
- Calibration checks
- Data diversity

